

Student Attendance Management System

System Description

Purpose

A C++ application that digitally tracks student presence across courses, replacing error-prone paper-based methods.

The Problem It Solves

Manual attendance is slow, prone to errors, and difficult to summarise. This system automates the entire workflow.

Core Features

- Add and manage students
- Assign instructors to courses
- Mark daily attendance
- Generate formatted reports

Main Classes

Seven classes form the backbone of the system, each with a clear, single responsibility.



Person

Base class for name and contact info



Student

Extends Person with a unique ID



Instructor

Extends Person with teaching duties



Course

Holds students, instructor, and records



AttendanceRecord

Stores presence status and date



AttendanceManager

Central controller for all logic



ReportHelper

Formats and outputs summaries

Class Responsibilities

Person

Stores and retrieves name and contact details. Base class for all people in the system.

Student

Extends Person; holds a unique student ID and displays enrolment info.

Instructor

Extends Person; manages assigned courses and teaching responsibilities.

Course

Maintains lists of enrolled students, a linked instructor, and attendance records.

AttendanceRecord

Logs a student's presence or absence alongside the session date.

AttendanceManager

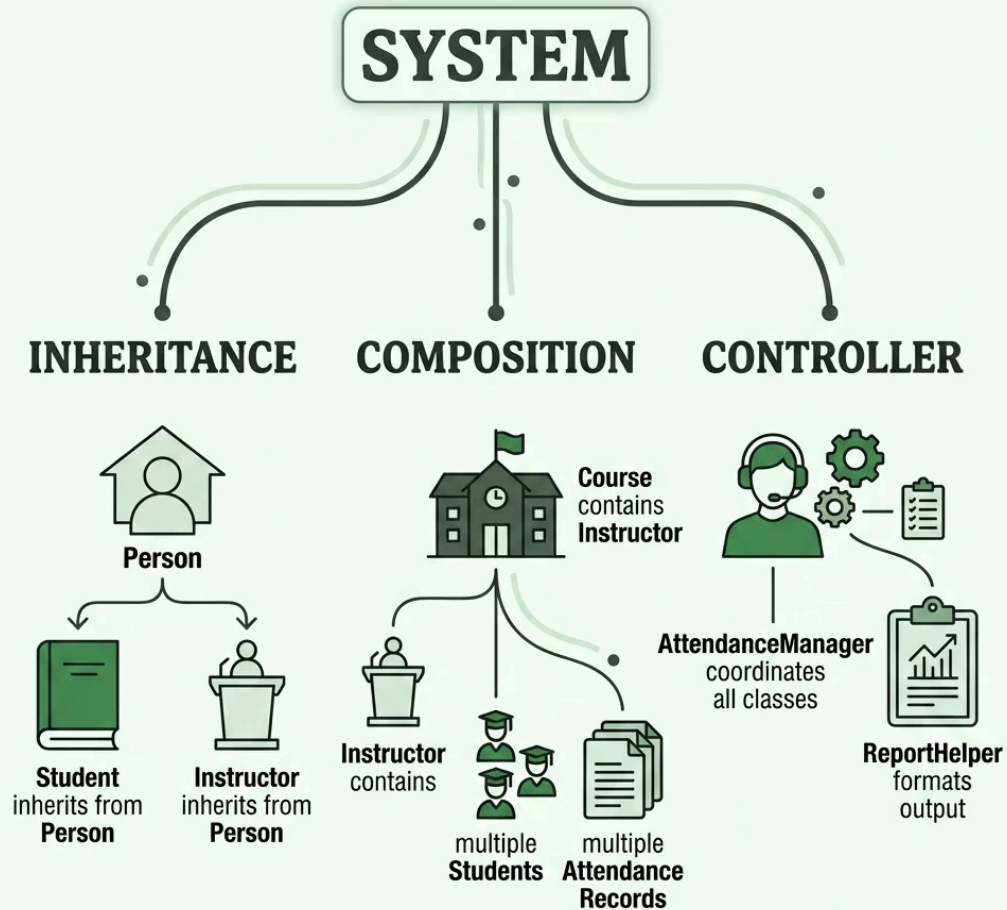
Orchestrates system flow: finds records, marks attendance, and triggers reports.

ReportHelper

Formats raw attendance data into readable, printed summaries.

 The **AttendanceManager** acts as the controller – all user actions pass through it.

Class Relationships & System Flow



Inheritance

Student and **Instructor** both extend **Person**, reusing name and contact attributes.

Composition

Course owns an **Instructor**, a list of **Students**, and **AttendanceRecords** – if the **Course** is removed, its records go with it.

System Flow

User selects a menu option → **AttendanceManager** processes the request → relevant classes are updated → **ReportHelper** outputs the result.